

TO: Executive Management & Rail Freight Operations Steering Committee

FROM: Technical Lead & Digital Engineering Team

CLIENT REF: Review of 'Cargo Trace – Nigerian Rail Freight Operations System'

CLASSIFICATION: Formal Memorandum of Operational Compliance

EXECUTIVE STATEMENT OF HARMONIZATION

Full Alignment with Nigerian Railway Corporation (NRC) Operating Standards

This official memorandum confirms the comprehensive completion, line-by-line harmonization, and live deployment of all operational corrections, codes, and functional refinements specified in your review of 'Cargo Trace – Nigerian Rail freight operations system'. Every single requirement—ranging from strict wagon code allocations, exclusive single-partner cement operations, dual track gauge physical segregation, dynamic station population, down to reactive universal user governance—is now actively functioning in production.

1. OPERATIONAL COMPLIANCE & IMPLEMENTATION MATRIX

Operational Domain	Client Specification & NRC Standard	Live Platform Implementation	Verification Point
01. Workflow & Consignment	Mandatory admission via formal Consignment Note declaring cargo, volume, origin, destination, and tariff request.	Official NRC-compliant Consignment Note in Customer Portal directly feeding Deal Desk & Marshalling Yard.	Customer Portal /customer
02. Wagon Classification	Carriage governed by exact NRC rolling stock codes: PXG, CBX, CBX(HS), OTW, ZGX, CHW(O) .	Generic wagons purged. All sidings, waybills, and deals strictly enforce specialized wagon codes and payload limits.	Deal Desk & Siding All wagon registries
03. Commercial Exclusivity	All competitor cement brands expunged. HBM is the sole cement manufacturing and rail partner.	Competitor mentions permanently removed. HBM operates dedicated 920 MT bulk block trains (23 PXG box wagons).	Trip TRP-8841 Papalanto → Moniya
04. Gauge Distinction	Clear operational distinction between Standard Gauge (1,435mm) and Narrow Gauge (1,067mm) .	Dual gauge badges on all deals. Route locks enforce physical network segregation between modern and legacy tracks.	Admin Deal Desk Corridor Badges
05. Production Cleanliness	Purge all cluttered test trips and trim platform to strictly two canonical production trips .	Storage migration (v7) executed. Exactly 2 trips live: TRP-8841 (HBM bulk cement) and TRP-9921 (APMT intermodal).	Live Fleet Tracker 2 Trips Active
06. Dynamic Terminals	Station and Moniya tabs must not be static mockups; they must populate trip-by-trip from train arrivals.	13-Column Terminal Information Sheet & Moniya Container Yard dynamically ingest arriving wagons and ISO containers.	Moniya Terminal Siding & Stacking Yard
07. User Governance	User Directory must be functional. Changing CEO name must instantly cascade to reports, invoices, and PDFs.	Reactive State Engine propagates user updates across deal desks, executive sign-off blocks, and exported PDFs.	User Directory & Reports /reports

2. IN-DEPTH TECHNICAL & OPERATIONAL ARCHITECTURE

Pillar I: NRC Rolling Stock Classification & Dynamic Consignment Note

Freight operations begin at the **Customer Freight Portal** where consignors declare their cargo using official NRC consignment documentation. The system enforces strict mechanical suitability—cargo is programmatically locked to certified wagon types:

- **PXG / CGs (Covered Box Wagons):** Engineered exclusively for 50kg bagged cement (HBM) and general bagged freight. Equipped with weatherproof double sliding doors and tamper-evident seal locks.
- **CBX (Flat-Bed Wagons):** Equipped with international twist-locks for 20ft and 40ft ISO shipping containers (APMT), structural steel pipes, machinery, vehicle chassis, and heavy wire coils.
- **CBX (HS) (High-Sided Flat-Beds):** Optimized for heavy industrial jumbo bags and bulk mineral aggregates requiring lateral containment.
- **OTW (Oil Tank Wagons):** Dedicated bottom-discharge pressurized vessels for refined petroleum products and liquid chemicals.
- **ZGX (Open-Top Discharging):** Pneumatic side-discharging wagons designed for coal, gypsum, clinker, and mining ores.
- **CHW(O) (Hopper Wagons):** Top-loading, center/bottom-discharge hopper fleet dedicated to track maintenance and ballast distribution.

Pillar II: Commercial Exclusivity (HBM) & Physical Track Gauge Segregation

- 1. Sole Cement Partnership:** In strict accordance with directive, all competitor cement references have been purged. **HBM** is configured as the exclusive cement manufacturing and logistics partner. The standard operational model runs 920 Metric Tonne block trains (23 PXG box wagons carrying 18,400 bags) along the Papalanto → Moniya heavy freight corridor.
- 2. Dual Track Gauge Physics:** The Nigerian rail network features two incompatible track gauges, both represented with distinct visual signifiers and operational constraints:
- **Standard Gauge (1,435mm) [Cyan Badge]:** High-capacity modern rail accommodating 25-tonne axle loads on the Lagos (Apapa/Ebute Metta) → Ibadan (Moniya) corridor.
 - **Narrow Gauge (1,067mm) [Amber Badge]:** Heritage 3ft 6in corridor supporting specialized regional freight and inland sidings.

Pillar III: Production Cleanliness & Trip-to-Terminal Synchronization

All legacy mock trips have been expunged. The platform currently maintains strictly **two canonical operational trains** that serve as the single source of truth:

TRIP 1: TRP-8841 (In-Transit Heavy Freight) • Consignor: HBM Cement (Sole Partner) • Route: Papalanto Siding (PAPA) → Moniya Terminal (MONI) • Consignment: 920 MT (18,400 Bags, 50kg Bagged Cement) • Formation: 23 Covered Box Wagons (PXG-4401 to PXG-4423) • Crew: Lead Engineer Babatunde Adeleke Gauge: Standard (1,435mm)	TRIP 2: TRP-9921 (Arrived Intermodal Freight) • Consignor: APMT Container Logistics • Route: Apapa Port Complex (APT) → Moniya Terminal (MONI) • Consignment: 350 MT (Import ISO Containers) • Formation: 10 Intermodal Flat-Beds (CBX-8101 to CBX-8110) • Crew: Unloading Officer Musa Ibrahim Gauge: Standard (1,435mm)
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Terminal & Siding Reactivity: Neither the Moniya Container Yard nor the Terminal Information Sheets rely on hardcoded prototypes. When TRP-9921 arrived at Moniya, its 10 CBX flatbeds populated the station siding ledger, and its ISO containers (MSKU-948210-4, APMT-310492-1, etc.) populated the 3D stacking yard bays (Bay A-01 to A-10). Selecting Papalanto immediately reveals the 23 PXG wagons loaded with HBM cement.

Pillar IV: Universal Reactive User Governance & Signatory Synchronization

To ensure institutional compliance during executive audits, the **User Directory & Account Provisioning** system has been re-architected with reactive synchronization. Hardcoded signatory names have been completely eliminated. The centralized StateEngine.getSignatory(role) resolver guarantees that modifying any executive name (e.g. Chief Executive Officer) immediately propagates across:

- **Executive Certification Cards:** The sign-off credentials displayed in the Admin Portal.
- **Official Commercial Invoices:** Signature blocks for Head of Finance and Head of Operations.
- **Automated Consignment Manifests & Gate Passes:** Station Master and Lead Conductor authorization tags.
- **Quarterly Executive Reports (/reports):** The formal sign-off certificate at the conclusion of financial summaries.
- **Exported & Printed PDF Documents:** All server-side and client-side generated documentation.

3. EXECUTIVE 3-MINUTE VERIFICATION PROTOCOL

To personally verify these implementations, please execute the following rapid validation sequence in your web browser:

Step	Action to Perform	Expected Concrete Observation
1	Navigate to Admin Portal → Live Fleet & Waybills	Notice strictly two active trips: TRP-8841 (HBM Cement, 23 PXG wagons) and TRP-9921 (APMT Containers, 10 CBX flatbeds). All legacy test data is gone.
2	Open Terminal Information Sheet and toggle stations	Notice the 13-column NRC sheet. Selecting Papalanto displays the 23 PXG box wagons. Selecting Moniya displays the 10 CBX flatbed wagons from TRP-9921.
3	Open Moniya Container Terminal tab	Notice true ISO container codes stacked in Bays A-01 through A-10 ingested from the arrived APMT train.
4	Inspect Deal Desk cards	Observe prominent Standard Gauge (1,435mm) cyan badges and Narrow Gauge (1,067mm) amber badges on operational corridors.
5	Go to User Directory , click Edit on the CEO row, change the name to a test name (e.g. <i>Alhaji Aliko Mohammed</i>), and click Save	Immediately observe the new CEO name reflected on the Executive Sign-off card, in official invoice modals, on /reports, and on exported PDFs.

4. FORMAL TECHNICAL & OPERATIONAL ENDORSEMENT

PREPARED BY:	REVIEWED & CONFIRMED:	APPROVED FOR SUBMISSION:
Lead Digital Systems Architect Bueno Rail Systems Engineering Certified NRC Digital Systems Integration	Head of Rail Freight Operations Bueno Logistics Freight Control Center Operational Compliance & Marshalling Division	Chief Executive Officer Bueno Logistics Limited Executive Directorate